TOPOSCOUT

GNSS SURVEY PLANNING SYSTEM

SITE INTELLIGENCE REPORT

REPORT SG-GHPEA8
Generated July 13, 2026
Data current as of report date

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	**66 ft** NAVD88 · MSL	**CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)**	**Zone X** 06055C0510FF	**13.0° E** WMM 2026

APN **038050009000** | **ZONING** **AP** | **ACRES** **0.481 ac**

BEST GNSS WINDOW **Mon, Jul 13 · 6:00 AM – 10:00 AM** | **PDOP 0.99 · 34 sv** | **NEAREST CORS** **P261 · 14.1 mi ✓**

REPORT CONTENTS

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	**66 ft**	**13.0° E**	**X**
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX
4.7

AVG KP
2.4

SOLAR FLUX F10.7
138 sfu

TEC (VTEC) ⚠ DEGRADED
12.4 TECU
-5.3 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change — verify before fieldwork

🛰 GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window

PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast | 💨 12 mph | **GO**

High 92°F | Low 61°F | Wind max 12 mph | Gusts 13 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

🌘 Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 11.5 mph (gusts 13.4 mph) | Precip: 0" | Temp: 61.3°–92.3°F

🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 11.5/13.4 mph S

☀ Sunrise: **5:56 AM** ☾ Sunset: **8:33 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph ⚡=turbulence	DIR
5 AM	2/2 mph	WNW
6 AM	1/1 mph	NE
7 AM	4/5 mph	ESE
8 AM	5/7 mph	SSE
9 AM	5/5 mph	SE
10 AM	3/3 mph	S
11 AM	0/2 mph	SE
12 PM	7/7 mph	S
1 PM	7/8 mph	S
2 PM	9/10 mph	S
3 PM	7/9 mph	S
4 PM	12/13 mph	SSW
5 PM	9/12 mph	SSW
6 PM	9/12 mph	S
7 PM	9/11 mph	S
8 PM	7/11 mph	SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

AI FIELD ANALYSIS — CLAUDE HAIKU 4.5

FIELD PLANNING REPORT
GNSS SURVEY OPERATIONS — NAPA COUNTY, CA
Generated: Monday, July 13, 2026

==
DAY RANKING & GO/NO-GO VERDICT
==

VERDICT: GO

REASONING: Excellent PDOP geometry (avg 0.99, best window 0.94 with 34-satellite constellation) combined with acceptable weather (overcast, dry, light winds) and moderate ionospheric conditions. Single unhealthy GPS satellite (PRN 11) does not degrade multi-constellation availability. Field operations are supported.

==
RECOMMENDED SURVEY WINDOW
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PRIMARY WINDOW: 06:00 AM to 10:00 AM local time (Pacific Daylight Time)
— PDOP: 0.94 (optimal geometry)
— Satellite count: 34 (GPS+GLONASS+Galileo+BeiDou)
— Duration: 4 hours continuous
— Conditions: Overcast; wind max 12 mph; zero precipitation forecast

RATIONALE: This window provides the tightest DOP geometry of the day. Morning overcast will not obstruct GNSS signals. Wind and precip forecasts present no risk.

==
BEST 4-HOUR STATIC OCCUPATION
==

TIME: 06:00 AM – 10:00 AM (Monday, July 13)
— Continuous static: 4 hours
— Expected PDOP: 0.94
— Elevation mask: 13° (terrain-based; low multipath risk in dwarf scrub environment)
— Ionospheric correction: Use network RTK or post-processing with ionospheric model; moderate Kp activity (avg 2.4) may introduce 5–15 cm delays on long baselines. Consider extended processing window or dual-frequency receivers for precise work.

OCCUPATION GUIDANCE: Moderate geomagnetic activity and TEC depression (-5.3 TECU) suggest ionospheric delay on the low side of normal. Four-hour occupation is sufficient for cm-level static positioning. If sub-cm accuracy required, post-process with global ionosphere map (GIM) corrections.

==
DRONE OPERATIONS ASSESSMENT
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DRONE STATUS: APPROVED for operations during 06:00–10:00 AM window

CONDITIONS SUMMARY:
— Wind: 12 mph max (acceptable for most small UAS; verify airframe limits)
— Visibility: Overcast (adequate for visual line-of-sight operations and photogrammetry)
— Precipitation: None forecast
— AQI: 45 (Good) — no air quality restrictions

CAUTION: Overcast conditions reduce contrast for orthophoto mosaic processing. Plan sufficient ground control points (GCPs) for photogrammetry tie-in if image-based georeferencing required.

==
FIELD SAFETY WARNINGS
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1. SATELLITE HEALTH: GPS PRN 11 currently unhealthy (UNUSUFN). Receivers will auto-exclude. Constellation remains robust; multi-constellation mitigation adequate.

2. MODERATE GEOMAGNETIC ACTIVITY: Kp 4.7 peak (moderate range). Minimal risk to GNSS operations; standard practice applies. No special precautions needed.

3. HEAT & HYDRATION: High temperature forecast (92°F). Crew should carry adequate water and take shade breaks during extended static occupation.

4. WILDFIRE STATUS: No active fires detected in reporting range. All-clear. Maintain situational awareness if conditions change.

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SUMMARY
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Monday, July 13, 2026 presents excellent conditions for GNSS survey work at the Napa County site (38.3459, -122.3047). The 06:00–10:00 AM window delivers optimal satellite geometry (PDOP 0.94, 34 satellites) with supportive weather (overcast, calm winds, dry). Moderate ionospheric activity is manageable with standard protocols. A 4-hour continuous static occupation during this window will satisfy most accuracy requirements; post-processing with ionospheric corrections is recommended for high-precision applications. Drone operations are cleared. Crew should manage heat exposure and monitor wildfire status. Proceed with field work as planned.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS — WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	**P261**	HUNTERHILLCN2004	14.1 mi
NGS ↗	**P198**	PETALUMAIRCN2004	17.5 mi
NGS ↗	**OHLN**	OHLONE PARK	23.5 mi
NGS ↗	**P196**	MEACHUMLFLCN2006	24.0 mi
NGS ↗	**CASR**	SANTA ROSA CA	24.9 mi
NGS ↗	**P267**	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS — WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗	**JT9010**	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9018**	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9009**	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9017**	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9034**	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
Datasheet ↗	**JT9012**	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
Datasheet ↗	**JT9043**	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X — AREA OF MINIMAL FLOOD HAZARD

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

DATA COLLECTOR IMPORT — NGS BENCHMARKS

↓ Download .txt File

```
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API

PID      NAME                    LAT          LON          ELEV_FT  ORDER

JT9010   13                       38.348333    -122.302778      62.7
JT9018   13 E                     38.348333    -122.302778      59.0
JT9009   13                       38.346667    -122.301111      58.7
JT9017   13 D                     38.345000    -122.301111      58.7
JT9034   20 A                     38.345000    -122.309444      68.0
JT9012   13                       38.351667    -122.304444      62.5
JT9043   25 A                     38.340000    -122.302778      61.4
JT9016   13 D                     38.341667    -122.297778      59.7
```

USGS STREAM GAUGES — within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.35 ft	3.78 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.83 ft	2.6 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.40 ft	2.67 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.86 ft	582 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	11.74 ft	6,820 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS — within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.01 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.91 ft (-9 ft)	706 ft	36K af	**67%**
Folsom Lake	66.5 mi	447.94 ft (-18 ft)	466 ft	785K af	**80%**
Lake Oroville	93.4 mi	860.82 ft (-39 ft)	900 ft	2,854K af	**81%**
New Melones Lake	100.1 mi	1,019.98 ft (-68 ft)	1,088 ft	1,662K af	**69%**
Black Butte Lake	101.2 mi	—	228 ft	74K af	**54%**

USBR RESERVOIRS (Bureau of Reclamation) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP

ARMY CORPS RESERVOIRS (USACE) — within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	**100%**
Camanche	89 mi	230.3 ft	221.86 ft	—	**104%**
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	**101%**
Farmington Dam-Pool	99 mi	125.5 ft	117.09 ft	—	**107%**
Black Butte-Pool	101 mi	456.9 ft	458.36 ft	—	**100%**
Don Pedro	137 mi	—	N/R	—	**—**

USGS GROUNDWATER WELLS — within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Coastal Range/Coast Ranges Formation aquifer in this area consists of fractured Tertiary sedimentary and volcanic rocks that accumulated in marine and terrestrial depositional environments. Water availability depends on fracture connectivity and seasonal recharge from local precipitation; transmissivity is typically moderate, and water tables show both seasonal and long-term variability tied to regional pumping and drought cycles.

FORMATION

Name	*Coastal Range Formation (Tertiary)*
Age / Rock	*Tertiary age; primarily siliciclastic and volcanic sedimentary rocks with minor interbedded volcanic flows.*
Extent	*800–1200 sq mi (est. USGS/DWR)*

DEPTH & RECHARGE

Depth Range	*51–295 ft (5–30 floors)*
Typical Depth	*100–150 ft (10–15 floors)*
Recharge Source	*Precipitation on upland outcrops and hillslope lateral flow through fractured bedrock*
Transit Time	*20–50 years to 150 ft depth (est. USGS/DWR)*

WATER QUALITY

pH	*est. 6.8–7.4 (est. USGS/DWR)*
Hardness	*est. 180–350 mg/L CaCO3 (est. USGS/DWR)*
TDS	*est. 280–550 mg/L (est. USGS/DWR)*
Notes	*No sample data — typical for formation. Generally suitable for domestic use; regional arsenic occurrence in volcanic facies not ruled out.*

TREND

Trend	*Long-term decline from 1977 to 2022: water levels dropped 9.9–76 ft over 45 years; steepest decline at deeper wells, consistent with regional groundwater stress.*
Source	*USGS NWIS historical well data at site and regional Napa County aquifer studies*
Seasonal Swing	*8–20 ft seasonal swing (est. USGS/DWR for Coastal Range Formation)*

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The site is located in the North Coast Ranges physiographic province with well-drained soils underlain by the N100CACSTL aquifer system, which is characterized by consolidated bedrock and alluvial deposits that have not experienced significant compaction. The area is not within any SGMA-designated critically overdrafted groundwater basin, indicating that groundwater extraction has remained within sustainable yield parameters and has not induced the clay layer consolidation typical of subsidence-prone regions. Surveyors working at this location can proceed with standard elevation measurement protocols without the need for subsidence-related corrections or benchmark verification adjustments.

SUBSIDENCE ASSESSMENT

Susceptibility	*Low*
Primary Mechanism	*None - consolidated terrain*
Annual Rate	*<1 mm/yr (est. USGS)*
Historic Loss	*None documented*
Aquifer Compaction	*None - aquifer not over-pumped*
SGMA Status	*Not applicable - not in SGMA basin*

BENCHMARK IMPLICATIONS

Monument Risk	*Low - monuments stable*
Elevation Validity	*Good - no active subsidence*
GNSS Recommendation	*Standard benchmark verification adequate*

SURVEY ACTION

Required Action	*No special action required*
Dataset	*CA DWR SGMA Bulletin 118 2024.1*
Coverage	*California critically overdrafted basins only - full US coverage pending*

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	**12 in minimum**

Est. from IECC Zone 3C — AFI data unavailable
Verify with local building official before construction

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with minimal temperature swings, so your primary envelope concern is managing moisture and wind rather than extreme heat or cold—prioritize continuous air sealing and durable cladding that can handle coastal salt spray and the 92 mph winds. Maximize south and west-facing glazing for passive solar gain during cooler months, but shade or minimize glass on the north and east facades where sea breezes bring fog and marine layer clouds that reduce solar benefit. Your best passive move for this zone is designing deep overhangs (3–4 feet) on south-facing windows to block summer sun while allowing winter sun to penetrate, which works year-round here since you'll rarely need heating but frequently need cooling modulation and glare control.

⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in northern California anticipates conventional spread footing construction on well-drained gravelly loam soils with moderate wind exposure and low snow loading. Seismic parameters are currently unavailable (null values) and may warrant immediate clarification with the local building official and a licensed engineer before design advancement. Frost depth of approximately 12 inches suggests standard foundation setback planning, though verification with local jurisdiction is recommended.

FOUNDATION

Type	*Conventional spread footing anticipated, subject to seismic data confirmation and geotechnical investigation*
Basis	*Well-drained gravelly loam (Coombs series) suggests favorable bearing; however, seismic design class (SDC) data is null and may warrant engineer consultation to confirm appropriate foundation detailing and depth*

STRUCTURAL IMPLICATIONS

Seismic	*Seismic parameters (Ss, S1, SDS, SD1, PGA) are null and require clarification—consult engineer and local building official to establish design category and anticipated seismic detailing*
Wind Exposure	*Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II; coastal location may warrant confirmation of exposure classification*
Wind	*92 mph 3-second gust wind speed anticipated—standard wind load design methodology recommended; consult engineer for final member and connection sizing*
Snow	*Ground snow load of 5.8 psf anticipated with winter wind factor W2=0.35; standard roof design likely adequate, though local snow event history review suggested*

INVESTIGATIONS & INSPECTIONS

Geotech Required	*Yes—geotechnical investigation recommended to confirm Site Class D assumption, bearing capacity, settlement potential, and seismic parameters for design basis*
Liquefaction Screen	*Screening recommended—groundwater depth and soil saturation status should be evaluated; well-drained gravelly soils suggest lower risk, but confirmation advised*
Special Inspections	*May be required depending on final SDC determination—consult engineer and building official once seismic parameters are established*
Code Edition	*ASCE 7-22 / IBC 2021 (planning reference only)*
Site Class Note	*Site Class D assumed for planning purposes based on soil description; licensed geotechnical investigation strongly recommended to confirm classification and provide seismic and foundation design parameters*

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	**SERIES:** Coombs
DRAINAGE CLASS: Well drained	**HYDRO GROUP:** C
SOIL ORDER: Alfisols	**SUBORDER:** Xeralfs

⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf suborder) formed in alluvial and colluvial materials derived from mixed rock sources in the California coast ranges and interior valleys. This soil exhibits moderate clay accumulation in the subsoil typical of Alfisols and maintains good drainage on these 2–5% slopes. A surveyor or developer should expect a gravelly upper profile transitioning to a more clayey substratum, with generally stable conditions suitable for most conventional development provided drainage patterns are maintained and cut/fill work accounts for the gravel content.

Alluvial and colluvial materials from mixed sources; foothill and valley floor landscape positions; mixed oak-grassland vegetation.

DRAINAGE

Class	*Well drained*
Seasonal Saturation	*None within 60 inches*
Flooding Risk	*Minimal on 2–5% slopes*
Ponding Risk	*None*

ENGINEERING

Bearing Capacity	*3,500–4,500 psf (shallow foundations)*
Rating	*Good*
Shrink-Swell	*Moderate — clay accumulation in subsoil*
Frost Action	*Low*
Cut Suitability	*Good (gravel-rich upper profile)*
Fill Suitability	*Fair (subsoil clay — moisture conditioning required during compaction)*

SEPTIC

Suitability	*Conditional*
Notes	*Perc test required — clay loam to clay substratum may limit infiltration rates; evaluate seasonal water table and site-specific permeability*

HAZARDS

Excavation	*Moderate — 25–40% gravel content in upper profile; standard equipment adequate*
Expansive Clay	*None anticipated*
Erosion	*Low on 2–5% slopes; gully initiation possible in heavily disturbed or bare areas*
Other	*None*

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 13, 2026, 2:59 PM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

🛢 OIL & GAS WELLS California — within 10mi radius

8 total

0 active

7 plugged

By status:

7 Plugged **1** Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	**Plugged**	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	**Idle**	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	**Plugged**	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any actionable distance threshold. No active producers, injection wells, or other well types exist within relevant review distances.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

⛏ MINES & MINERAL DEPOSITS USGS MRDS — within ~35mi radius

200 total in search area

10 within 5 mi

108 within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	**Occurrence**	Sand/Gravel	1.6 mi
Basalt Rock Co	**Producer**	Clay	2 mi
Cicero Pumice Quarry	**Occurrence**	PUM	2.9 mi
Mount George King Pumice Deposit.	**Occurrence**	PUM	2.9 mi
Basalt Rock Co.	**Occurrence**	DIT	3.1 mi
Mt George Pumice Deposit	**Producer**	PUM	3.1 mi
Wilson Pumice Quarry	**Occurrence**	PUM	3.7 mi
Unknown	**Occurrence**	Mercury	4.4 mi
Silva Pumice Quarry	**Occurrence**	PUM	4.7 mi
Silva Pumice Quarry	**Producer**	PUM	5 mi
Walker Pumice Quarry	**Occurrence**	PUM	5.1 mi
Walker Pumice Quarry	**Producer**	PUM	5.1 mi
Mountain	**Occurrence**	Mercury	5.4 mi
Unknown	**Occurrence**	Mercury	7.6 mi
Unknown	**Occurrence**	Stone/Crushed	7.8 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Prospect	**Occurrence**	Chromium	8.8 mi
Unnamed Location	**Occurrence**	Chromium	9 mi
Unknown	**Occurrence**	Chromium	9.1 mi
Bella Oak	**Past Producer**	Mercury	9.4 mi
Unknown	**Occurrence**	MG	9.7 mi
Serres Pit	**Past Producer**	Sand/Gravel	10.4 mi
Unnamed Location	**Unknown**	Sand/Gravel	10.5 mi
Conn Valley	**Occurrence**	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Mineral Resource Site Planning Assessment Three active producers operate within 5 miles of the project site. Basalt Rock Co (2 mi) and Mt George Pumice Deposit (3.1 mi) are clay and pumice operations respectively, both generating heavy truck traffic, dust emissions, and potential noise from extraction/processing activities during operating hours. Silva Pumice Quarry (5 mi) similarly produces pumice with comparable operational impacts. No historical deposits warrant Phase 1 ESA consideration based on proximity thresholds.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 Endangered

6 Threatened

12 Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	*SCIENTIFIC NAME*	STATUS
California tiger Salamander	*Ambystoma californiense*	**Endangered**
Conservancy fairy shrimp	*Branchinecta conservatio*	**Endangered**
Contra Costa goldfields	*Lasthenia conjugens*	**Endangered**
Soft bird's-beak	*Cordylanthus mollis ssp. mollis*	**Endangered**
Suisun thistle	*Cirsium hydrophilum var. hydrophilum*	**Endangered**
Vernal pool tadpole shrimp	*Lepidurus packardi*	**Endangered**
Alameda whipsnake (=striped racer)	*Masticophis lateralis euryxanthus*	**Threatened**
California red-legged frog	*Rana draytonii*	**Threatened**
Delta smelt	*Hypomesus transpacificus*	**Threatened**
Northern spotted owl	*Strix occidentalis caurina*	**Threatened**
Vernal pool fairy shrimp	*Branchinecta lynchi*	**Threatened**
western snowy plover	*Charadrius nivosus nivosus*	**Threatened**

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Summary – 38.3459, -122.3047 **Section 7 Consultation Triggers:** Federal nexus activities (permits, funding, or approvals) triggering Section 7 consultation are likely given the concentration of 12 species with critical habitat within 25 miles, particularly for vernal pool-associated species (fairy shrimp, tadpole shrimp, Contra Costa goldfields) and California tiger salamander, which require specific habitat conditions commonly impacted by site development. **Most Planning-Relevant Species:** The California tiger salamander and vernal pool complex species warrant highest attention due to their restrictive habitat requirements and frequent conflict with land conversion; California red-legged frog and Alameda whipsnake are secondary priorities depending on site conditions. **Recommended Next Step:** Commission a Phase 1 biological survey and wetland delineation to determine presence/absence of potential vernal pools, seasonal wetlands, and terrestrial refugia; results will clarify whether formal consultation under Section 7 is necessary and inform project design constraints. Early coordination with the U.S. Fish and Wildlife Service regarding project scope is advisable to identify consultation pathways and potential avoidance strategies. --- *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
American black nightshade	*Solanum americanum*	2026-05-29
Bearded Clover	*Trifolium barbigerum*	2026-05-15
bird's foot cliffbrake	*Pellaea mucronata*	2026-04-28
Blow Wives	*Achyrachaena mollis*	2026-05-20
blue oak	*Quercus douglasii*	2026-05-09
Bolander's Phacelia	*Phacelia bolanderi*	2026-04-27
box elder	*Acer negundo*	2026-05-18
Braun's Giant Horsetail	*Equisetum telmateia braunii*	2026-06-22
California bay	*Umbellularia californica*	2026-06-24
California beeplant	*Scrophularia californica*	2026-05-20
California black oak	*Quercus kelloggii*	2026-05-15
California buckeye	*Aesculus californica*	2026-06-02
California Dutchman's Pipe	*Aristolochia californica*	2026-05-26
California milkwort	*Rhinotropis californica*	2026-05-15
California mugwort	*Artemisia douglasiana*	2026-06-02
California poppy	*Eschscholzia californica*	2026-06-24
California Wild Rose	*Rosa californica*	2026-05-09
Canyon liveforever	*Dudleya cymosa*	2026-05-30
-	*Castilleja densiflora densiflora*	2026-05-09
clay mariposa lily	*Calochortus argillosus*	2026-05-15
coast live oak	*Quercus agrifolia*	2026-05-13
coastal woodfern	*Dryopteris arguta*	2026-05-20
common selfheal	*Prunella vulgaris*	2026-06-22
Diogenes' lantern	*Calochortus amabilis*	2026-05-15
distant phacelia	*Phacelia distans*	2026-05-15
-	*Downingia concolor concolor*	2026-05-15
Dwarf Brodiaea	*Brodiaea terrestris*	2026-05-15
Elegant Clarkia	*Clarkia unguiculata*	2026-05-16

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
goldback fern	*Pentagramma triangularis*	2026-05-20
Ithuriel's Spear	*Triteleia laxa*	2026-06-15
maroonspot calicoflower	*Downingia concolor*	2026-05-15
narrowleaf milkweed	*Asclepias fascicularis*	2026-06-25
narrowleaf mule-ears	*Wyethia angustifolia*	2026-05-15
Narrowleaf Willow	*Salix exigua*	2026-06-24
needleleaf navarretia	*Navarretia intertexta*	2026-05-15
northern California black walnut	*Juglans hindsii*	2026-06-24
orange bush monkeyflower	*Diplacus aurantiacus*	2026-05-29
Oregon Ash	*Fraxinus latifolia*	2026-06-02
Pacific Bleeding Heart	*Dicentra formosa*	2026-05-04
Pacific madrone	*Arbutus menziesii*	2026-07-04
Pacific poison oak	*Toxicodendron diversilobum*	2026-06-02
pineapple-weed	*Matricaria discoidea*	2026-05-20
Redwood Sorrel	*Oxalis smalliana*	2026-05-28
small-headed clover	*Trifolium microcephalum*	2026-05-15
Superb Mariposa Lily	*Calochortus superbus*	2026-06-11
thimble clover	*Trifolium microdon*	2026-05-15
tomcat clover	*Trifolium willdenovii*	2026-05-15
Toyon	*Heteromeles arbutifolia*	2026-06-19
trailing blackberry	*Rubus ursinus*	2026-06-24
Tree Clover	*Trifolium ciliolatum*	2026-05-15
valley oak	*Quercus lobata*	2026-06-24
wavy-leafed soap plant	*Chlorogalum pomeridianum*	2026-05-20
western sycamore	*Platanus racemosa*	2026-05-17
white brodiaea	*Triteleia hyacinthina*	2026-05-15
white-tipped clover	*Trifolium variegatum*	2026-05-15
Winecup Clarkia	*Clarkia purpurea*	2026-05-15
Yellow Mariposa Lily	*Calochortus luteus*	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Site Ecology Assessment (38.3459, -122.3047) This site supports a **mixed oak woodland and chaparral community**, dominated by drought-tolerant species like blue oak, California black oak, and California buckeye, with scattered understory shrubs and herbaceous plants adapted to Mediterranean climates. The prevalence of clay-loving plants (clay mariposa lily, Castilleja) and moisture-indicator species (Braun's Giant Horsetail, box elder) suggests **seasonal wetness with summer drought stress**, typical of foothill areas with clay soils and intermittent water flow. Drainage design should account for **slow water infiltration through clay-rich soils and potential seasonal saturation in low-lying areas**, requiring grading that avoids disrupting natural drainage patterns and protects riparian or seepage zones where moisture-dependent species concentrate. These ecological conditions indicate the site likely experiences periodic water accumulation that supports specific plant communities, making careful stormwater management critical during any ground disturbance. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

636	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
Crotalus oreganus oreganus	636 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
Latrodectus hesperus	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
Dasymutilla aureola	91 obs
Dasymutilla californica	91 obs
Dasymutilla sackenii	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
Toxicodendron diversilobum	1100 obs
Datura wrightii	57 obs
Datura stramonium	57 obs
Datura ferox	57 obs
Conium maculatum	199 obs
Urtica gracilis	83 obs
Urtica gracilis holosericea	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Sonoma County, CA Area **Highest-Priority Hazards:** Northern Pacific rattlesnakes are your top concern with 636 observations in this region—they're active spring through fall and often stay concealed in brush or tall grass where survey crews walk. Poison oak is ubiquitous (1,439 observations) and causes severe dermatitis from any contact, including through clothing and tools. Black widow spiders (154 obs) hide in dark crevices and equipment, while several toxic plant species including deadly hemlock and Datura varieties present ingestion and skin contact risks. **Practical Avoidance:** Wear closed-toe boots, long pants, and long sleeves; use a walking stick to probe vegetation and alert snakes before you step. Learn to identify poison oak's three-leaflet pattern and avoid touching all vegetation—wash tools and clothing immediately after fieldwork. Check equipment, tents, and gloves before use. Keep antivenom information and nearest hospital locations readily available. **Seasonal Peak:** Late spring through early fall (May–October) is peak rattlesnake and insect activity; poison oak remains hazardous year-round but causes more exposures in warm months when crew members wear less protection. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	90	64	5	15	6
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (64 species)

COMMON NAME	*Scientific Name*	OBS
Wild Turkey	*Meleagris gallopavo*	7
California Quail	*Callipepla californica*	6
California Towhee	*Melozone crissalis*	6
Snowy Egret	*Egretta thula*	6
Western Bluebird	*Sialia mexicana*	5
Great Egret	*Ardea alba*	4
Red-tailed Hawk	*Buteo jamaicensis*	4
Black Phoebe	*Sayornis nigricans*	4
Northern Flicker	*Colaptes auratus*	3
Cooper's Hawk	*Astur cooperii*	3
Ash-throated Flycatcher	*Myiarchus cinerascens*	3
Western Screech-Owl	*Megascops kennicottii*	3
Turkey Vulture	*Cathartes aura*	3
Mallard	*Anas platyrhynchos*	3
Bushtit	*Psaltriparus minimus*	3

+ 49 additional species

MAMMALS (5 species)

COMMON NAME	*Scientific Name*	OBS
Coyote	*Canis latrans*	3
Black-tailed Jackrabbit	*Lepus californicus*	3
Mule Deer	*Odocoileus hemionus*	2
Eastern Fox Squirrel	*Sciurus niger*	1
Striped Skunk	*Mephitis mephitis*	1

REPTILES (15 species)

COMMON NAME	*Scientific Name*	OBS
Western Fence Lizard	*Sceloporus occidentalis*	10
Northern Pacific Rattlesnake	*Crotalus oreganus oreganus*	7
Pacific Gopher Snake	*Pituophis catenifer catenifer*	6
California King Snake	*Lampropeltis californiae*	3
Pond Slider	*Trachemys scripta*	2

COMMON NAME	*Scientific Name*	OBS
Western Skink	*Plestiodon skiltonianus*	2
Ring-necked Snake	*Diadophis punctatus*	1
Common Garter Snake	*Thamnophis sirtalis*	1
Western Yellow-bellied Racer	*Coluber constrictor mormon*	1
Gopher Snake	*Pituophis catenifer*	1
Pacific Ringneck Snake	*Diadophis punctatus amabilis*	1
Northwestern Pond Turtle	*Actinemys marmorata*	1
Common Sharp-tailed Snake	*Contia tenuis*	1
North American Racer	*Coluber constrictor*	1
Southern Alligator Lizard	*Elgaria multicarinata*	1

AMPHIBIANS (6 species)

COMMON NAME	*Scientific Name*	OBS
Pacific chorus frog	*Pseudacris regilla*	5
Western Toad	*Anaxyrus boreas*	3
American Bullfrog	*Lithobates catesbeianus*	3
Foothill Yellow-legged Frog	*Rana boylii*	3
Rough-skinned Newt	*Taricha granulosa*	1
California Giant Salamander	*Dicamptodon ensatus*	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a diverse wildlife community characteristic of foothill oak woodland and grassland transitional habitat with proximate riparian or wetland features. The bird assemblage reflects mixed open and scrub habitats with several species (egrets, pond turtle, aquatic amphibians) indicating water resources—likely seasonal creeks or ponds—while the robust reptile diversity (15 species) and presence of riparian-associated amphibians (Pacific chorus frog, California giant salamander, rough-skinned newt) suggest adequate moisture and structural complexity. Notable species of regulatory concern include the Foothill Yellow-legged Frog and California Giant Salamander, both of which require clean, cool-water breeding habitat and are sensitive to hydrologic alteration and sedimentation; these taxa indicate relatively intact stream or seepage habitat quality. The overall species richness and presence of native predators (raptors, snakes) alongside sensitive amphibians suggest moderate habitat integrity with low to moderate existing disturbance, though the presence of non-native species (American Bullfrog, Eastern Fox Squirrel, Pond Slider) indicates some prior land use impact; development review should prioritize protection of riparian corridors, seasonal wetlands, and upland connectivity to maintain this ecological function. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	← excellent
7/12	30.3°	98.8%	← excellent
8/12	33.7°	99.3%	← excellent
9/12	36.9°	99.8%	← annual optimal
10/12	39.8°	99.8%	← excellent
12/12	45°	99%	← excellent

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls

Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	⚠ verify
50ft	All year	4.8	4.6	7.3	⚠ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.6 hours of useful sunlight per day on average, which is genuinely excellent for solar panels—think of it like having a part-time job that pays consistently, even on cloudy days. Your roof's pitch and color are surprisingly powerful: a dark roof acts like a heat-absorbing sponge in summer (pulling in 301 BTU per hour per square foot), while a light-colored roof would cut that in half, so choosing white or light gray is like putting on sunscreen for your house. Here's the big gotcha most people miss—your west-facing windows are like an open oven door facing afternoon heat, bringing in over 100 BTU per hour in summer, which your air conditioning will have to work hard against, so think about shading those windows with curtains or an awning. The good news is that a large deciduous tree on your south side is almost free climate control: it'll shade your house for about 4 hours during summer heat, but then lose its leaves in winter to let that same sun warm your place when you need it—it's like nature's seasonal thermostat. Thermal mass (thick materials like concrete or adobe) is simply anything that soaks up heat slowly during the day and releases it slowly at night, evening out your temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

🧠 **SITE INTELLIGENCE SUMMARY** AI synthesis — Claude Sonnet

⚙ **Technical** 👤 **Client**

Technical Brief — for architects, engineers, and surveyors

STRUCTURAL DESIGN DRIVERS

Climate governs the structural envelope for this Napa County site. IECC Zone 3C (Marine) with an accumulated freezing index of 0°F-days eliminates frost-driven design cycles and winter precipitation loading as primary concerns; wind and snow loads are not quantified in the available data and should be confirmed through ASCE 7 site-specific analysis. The absence of seismic hazard data prevents determination of whether seismic or wind will dominate lateral design. Recommend immediate acquisition of site-specific seismic hazard parameters (PGA, spectral accelerations) and wind speed data (3-second gust, exposure category) to establish which governs member sizing and foundation moment capacity.

GEOTECHNICAL CONSIDERATIONS

The site is underlain by Coombs gravelly loam with 2 to 5 percent slopes, classified as NRCS Hydrologic Group C (well drained). This soil type will accommodate shallow foundations with standard design practices; Group C soils drain within 24 hours, reducing risk of ponding and frost heave—particularly relevant given the zero frost-days climate. The texture classification is incomplete in the dataset and must be confirmed via boring and laboratory analysis before final footing depth and bearing capacity are selected. Well-drained slopes at this gradient present no obvious mass-movement risk, but subsurface exploration should verify gravel and clay distributions, as gravelly loams can exhibit variable compressibility depending on fines content.

HAZARD AND SUBSIDENCE ASSESSMENT

No Quaternary faults are documented within the search radius, and no subsidence basin data apply to this location. These absences do not eliminate seismic risk—fault rupture and subsidence hazards must be reconfirmed through county geological records and USGS documentation specific to the site latitude and longitude. Oil and gas activity in the region is minimal (8 wells within 10 miles, none active or producing; nearest plugged well 7.2 miles away), reducing induced seismicity and ground-stability concerns from that source. Mining activity is present but distant: 3 active producers within 5 miles (two pumice quarries and one basalt operation at 2–5 miles) pose no immediate ground stability or vibration hazard to the site.

SITE CONTEXT AND CLIMATE ENVELOPE

The site lies in a marine-influenced climate zone with no seasonal frost and minimal thermal cycling—a favorable envelope for concrete durability and reduced need for cold-climate detailing. Napa County's mining context (200 MRDS deposits regionally, 3 active producers nearby) reflects regional geology rather than site-specific risk; however, ground stability and vibration monitoring may become relevant if future operators expand production within 2–3 miles. The marine climate classification typically correlates with moderate humidity and salt exposure risk if the site is near coast or valley areas—confirm distance to marine/saline influence sources.

BOTTOM LINE

Obtain site-specific seismic hazard parameters and wind speed data immediately, as these will determine whether lateral load governs structural system selection, and commission a Phase I geotech boring to confirm soil texture, bearing capacity, and footing depth before advancing schematic design.

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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

SITE SUMMARY

Coordinates	38.345860°N 122.304660°W
Elevation	66 ft NAVD88 (approx)
Township/Range	T6N, R4W, Mount Diablo Meridian
FEMA Zone	Zone X Panel 06055C0510FF
Nearest BM	JT9010 0.2mi 62.7ftft
Nearest CORS	P261 14.1mi
State Plane	CA 2 (0402)

TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

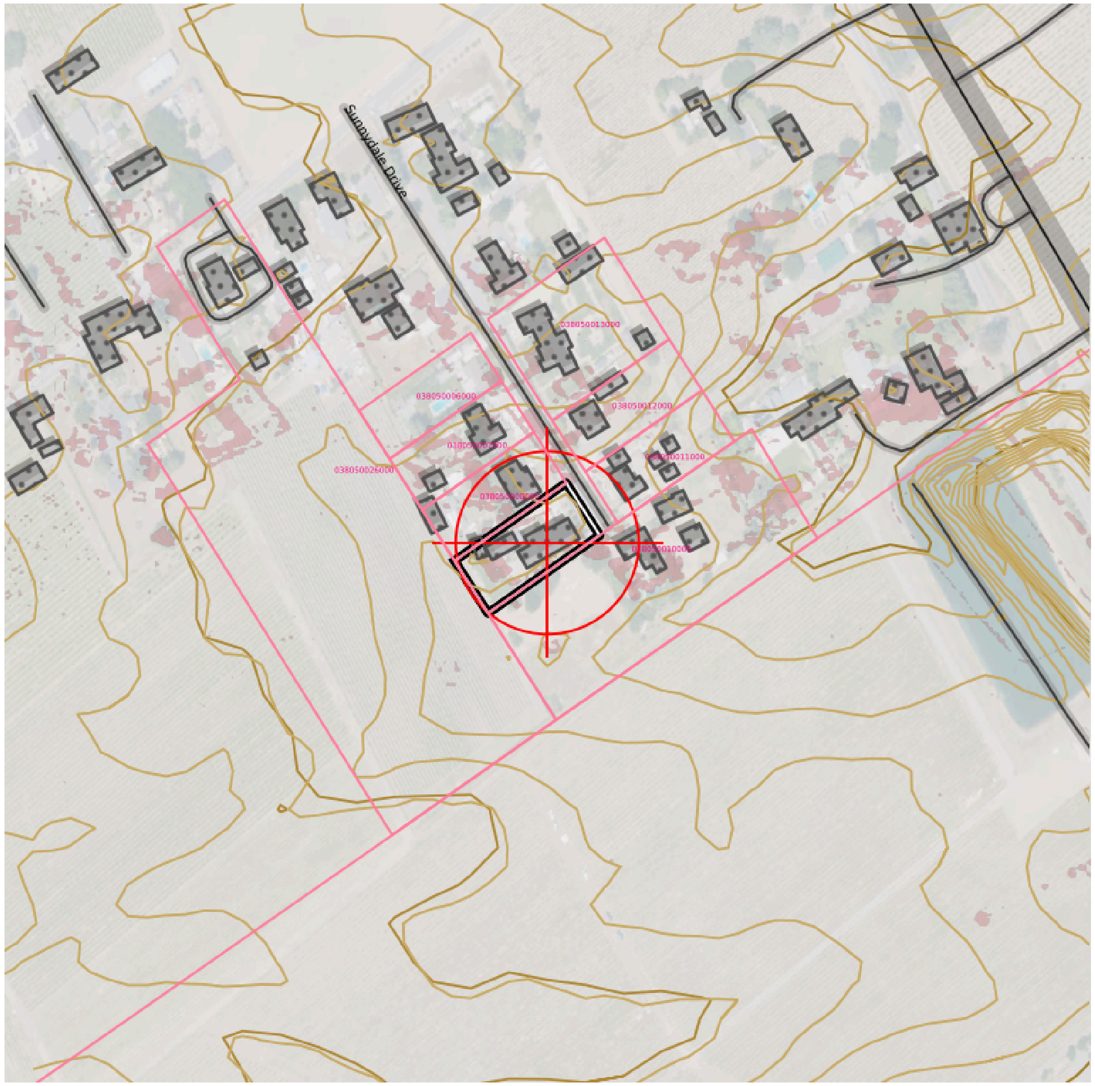
Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

TOPOSCOUT SITE DXF PREVIEW

45 CAD Layers · AutoCAD Compatible · State Plane Coordinates

Need the editable CAD file? Add the DXF at toposcout.com

TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

N

0 —— 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels ■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

SITE AERIAL — 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial

TOPOSCOUT SITE AERIAL IMAGERY

Source: Google Aerial

AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

⚡ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

BASE STATION & NETWORK

NOAA CORS Network – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

NGS CORS Station Map – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

IGS Station Monitor – https://www.igs.org/network
International GNSS Service – global base station health

POST-PROCESSING

NGS OPUS – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service – static GNSS processing

NGS OPUS-RS – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing – sessions as short as 15 minutes

CORS Data Download – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

REFERENCE & CALCULATIONS

NGS Datasheets – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

NGS Geoid Models – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

FEMA & FLOOD

FEMA Flood Map Service Center – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

DRONE OPERATIONS

FAA Part 107 — Commercial Rules – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model

Links verified: February 2026 · Check quarterly · TopoScout GNSS Planning System